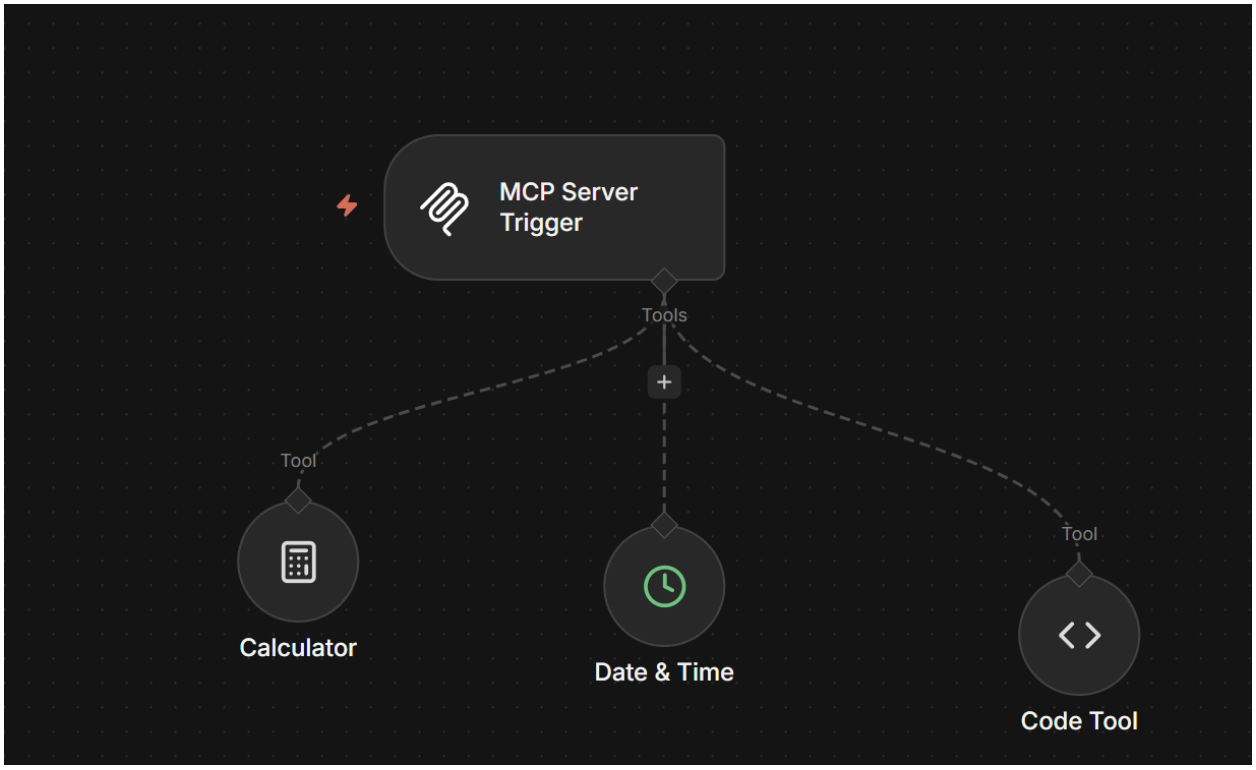
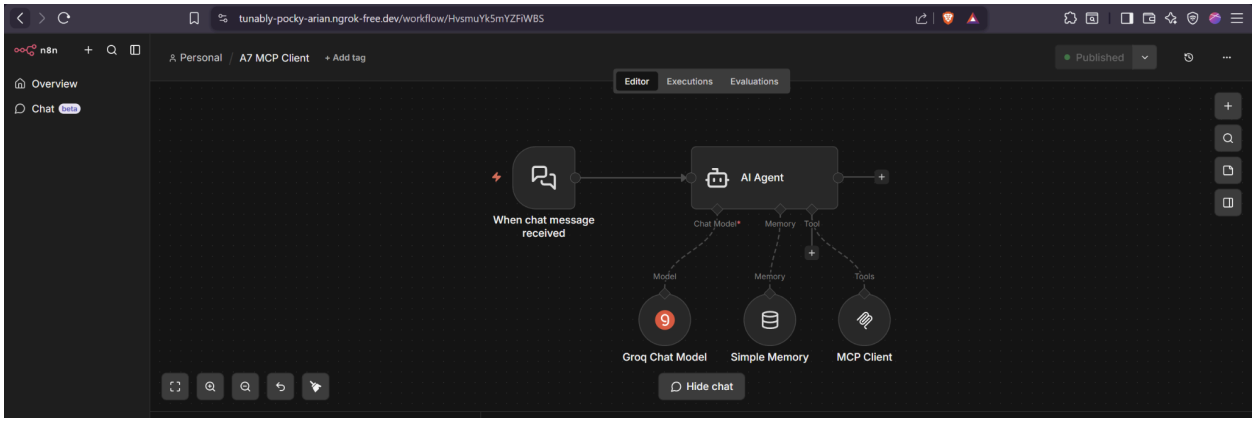
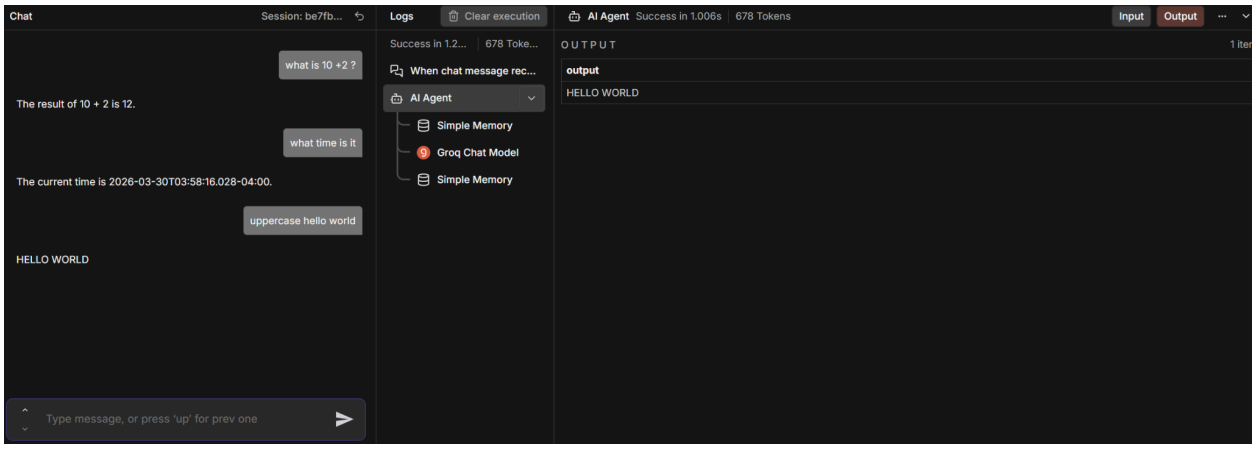
Assignment Explanation Report

Task 1: MCP Infrastructure & Server Setup

- **Server Deployment:** The n8n environment was successfully deployed locally using Docker, and an ngrok tunnel was utilized to expose the Production URL to the internet, allowing for external webhook and API communication.
- **MCP Server Workflow:** An n8n workflow was established to act as the MCP Server. This included an MCP Trigger connected to internal tools such as the Calculator and Date/Time tool.
- **AI Agent Client Configuration:** A separate AI Agent workflow was created utilizing the Groq Chat Model API. Context was maintained using a Simple Memory node. The MCP Client tool was connected to the MCP Server's URL, and the integration was successfully verified in the n8n chat interface by prompting the agent to perform basic arithmetic (e.g., 10 + 2 = 12) and retrieve the current date and time.

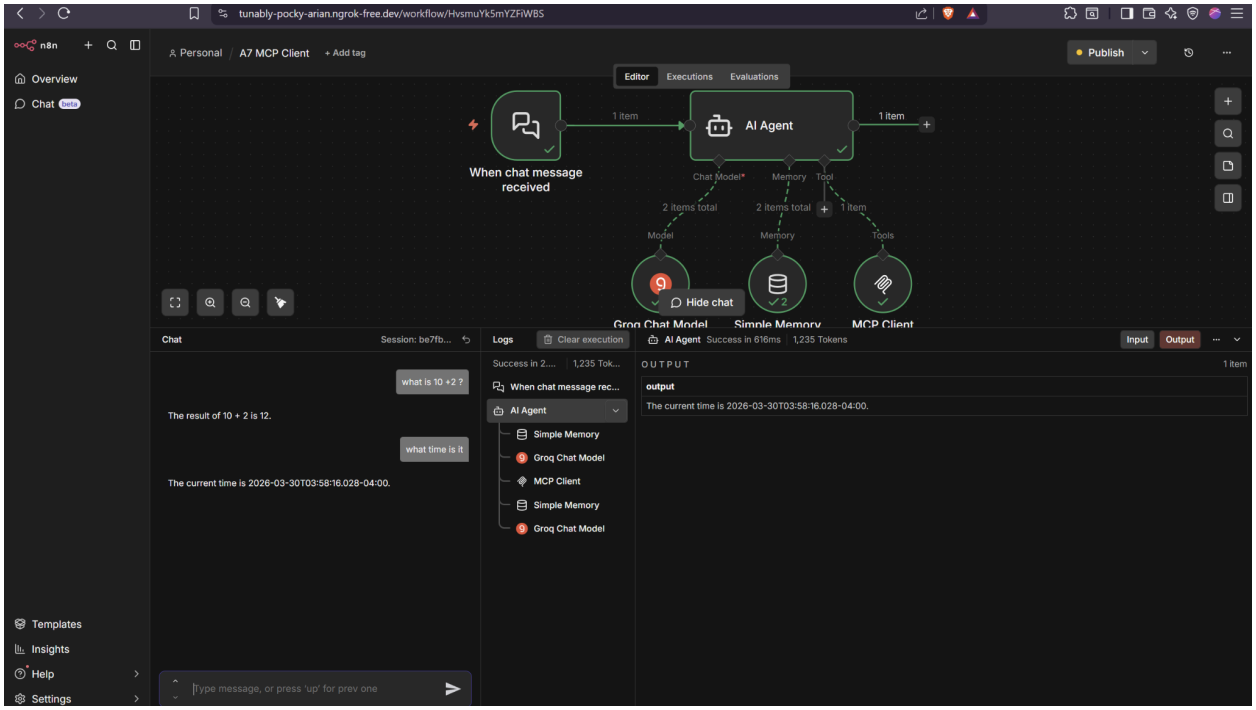






Task 2: Telegram & Google Calendar Integration

- **Telegram Bot Integration:** The AI Agent's chat trigger was replaced with a Telegram Trigger, enabling the agent to receive and respond to user messages directly through the Telegram messaging app.
- **Google Calendar Tool:** The Google Calendar tool was added to the AI Agent's toolset, granting it the necessary permissions to read, create, and manage calendar events.
- **Automated Project Scheduling:** Through the Telegram interface, a command was issued to the agent to schedule a project. The agent successfully processed the NLU request and created the four required events in Google Calendar:
    - 1st Phase: Literature Review
    - 2nd Phase: Project Proposal
    - 3rd Phase: Update Progress
    - 4th Phase: Final (Presentation)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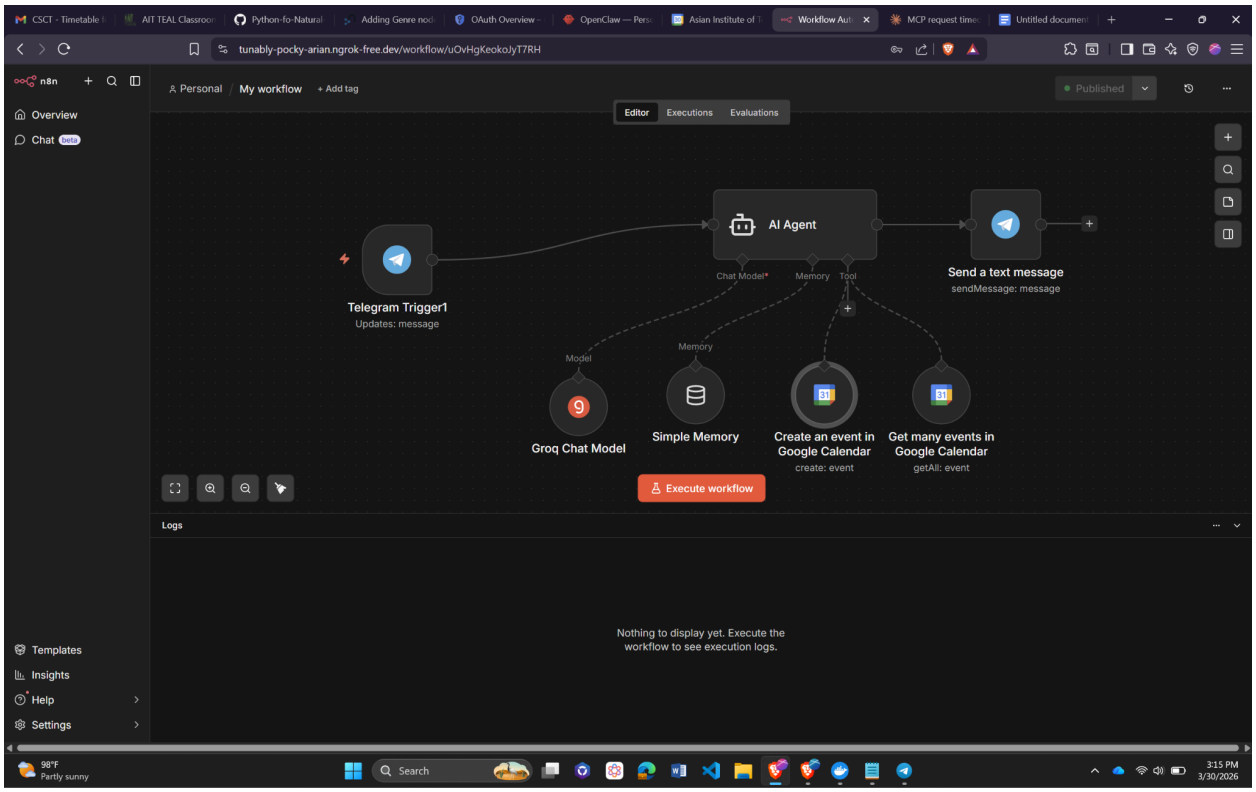
n8n
Personal / A7 MCP Client
+ Add tag
Publish
Overview
Chat beta
Editor
Executions
Evaluations
When chat message received
1 item
AI Agent
1 item
Chat Model*
Memory
Tool
2 items total
2 items total
1 item
Model
Memory
Tools
Groq Chat Model
Simple Memory
MCP Client
Hide chat
Chat
Session: be7fb...
what is 10 +2 ?
The result of 10 + 2 is 12.
what time is it
The current time is 2026-03-30T03:58:16.028-04:00.
Type message, or press 'up' for prev one
Logs
Clear execution
Success in 2.... 1,235 Tok...
When chat message rec...
AI Agent
Simple Memory
Groq Chat Model
MCP Client
Simple Memory
Groq Chat Model
AI Agent Success in 616ms 1,235 Tokens
Input
Output
OUTPUT
1 item
output
The current time is 2026-03-30T03:58:16.028-04:00.
Templates
Insights
Help
Settings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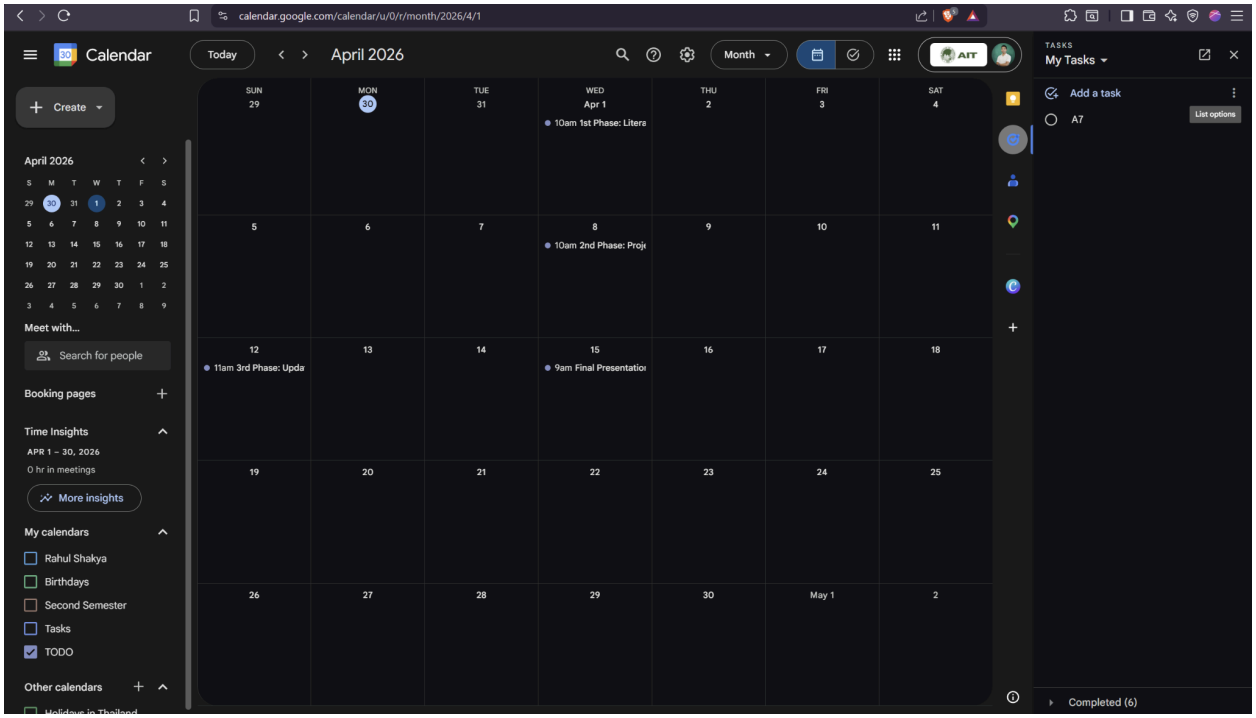
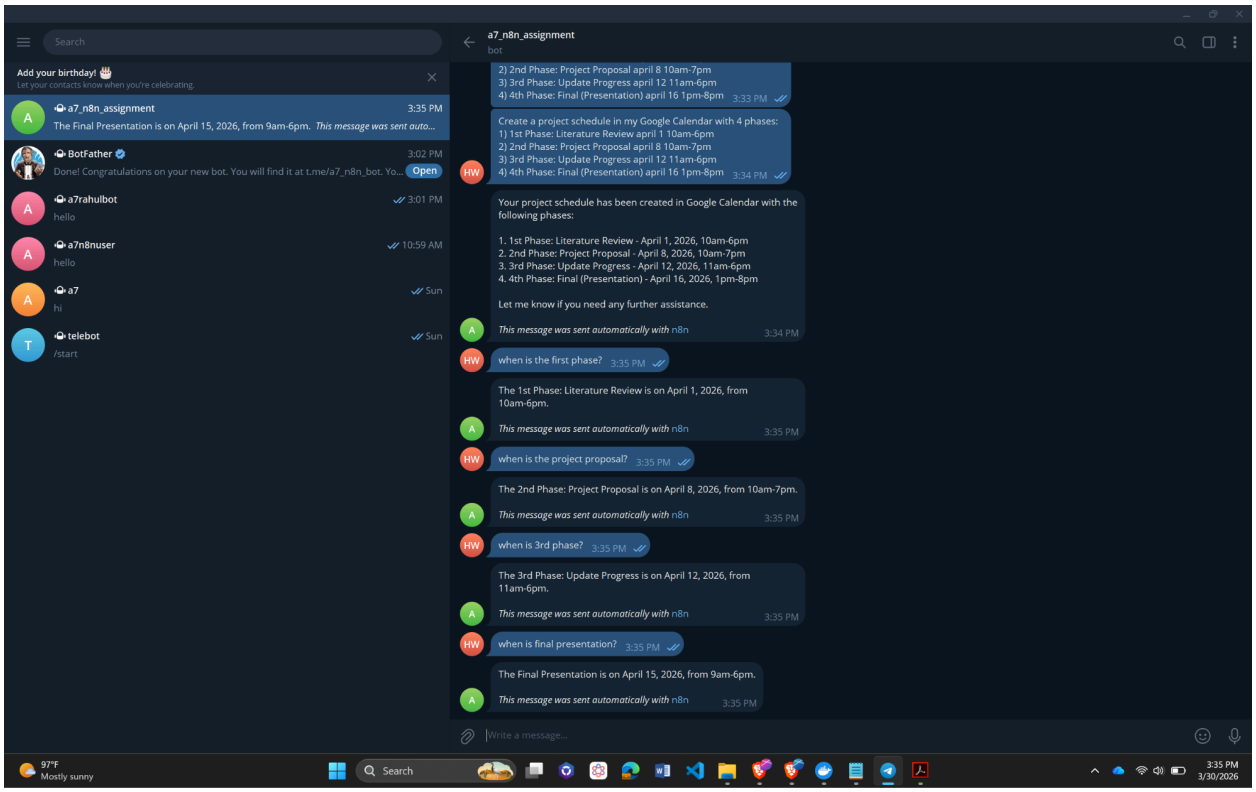
n8n
Personal / My workflow
+ Add tag
Published
Overview
Chat beta
Editor
Executions
Evaluations
Telegram Trigger1
Updates: message
AI Agent
Chat Model*
Memory
Tool
Send a text message
sendMessage: message
Model
Memory
Groq Chat Model
Simple Memory
Create an event in Google Calendar
create: event
Get many events in Google Calendar
getAll: event
Execute workflow
Logs
Nothing to display yet. Execute the workflow to see execution logs.
Templates
Insights
Help
Settings
98°F
Partly sunny
Search
3:15 PM
3/30/2026



Interaction Verification: The agent's capabilities were verified by querying the chatbot in Telegram about the specific dates of the scheduled project phases. The agent accurately recalled the scheduled times, and the events were visually confirmed to be populated within the Google Calendar interface.